One neuron

AI · Level 3 · Lesson 1



Today we will

- Work out a neuron’s output from its inputs, weights and bias.
- Explain weights as how much each input matters, and learning as changing them.
- Explain why one neuron can only draw one straight line between two answers.

Remember

- A neuron multiplies each input by its weight, adds a bias, and decides: yes if the total is above zero.
- Weights say how much each input matters. Training changes them a little at a time: that is learning.
- One neuron draws one straight line. Questions with yes in the middle need more neurons.

Be a human neuron

1. Warm-up (5 min): How do you decide whether to take an umbrella?
2. Teach (8 min): Inputs, weights, bias and the total, worked on the board.
3. Human neuron (14 min): Groups of four play the umbrella neuron.
4. Change a weight (7 min): Make one weight zero, then negative.
5. Wrap-up (6 min): The one-line limit, then the exit ticket.

Make input cards (1 on one side, 0 on the other) and weight cards (3, 2, 1) and a bias card (-1) for each group. Plan five days to call out, including one where the neuron gets it wrong.

Let’s discuss

1. Which input in our umbrella neuron should matter most, and why?
2. What would a weight of zero mean in real life?
3. Which decisions in school are “just right in the middle”?

Exit ticket

- Inputs 1, 0 and 1, weights 1, 3 and 2, bias -4: what is the total, and what does the neuron say?